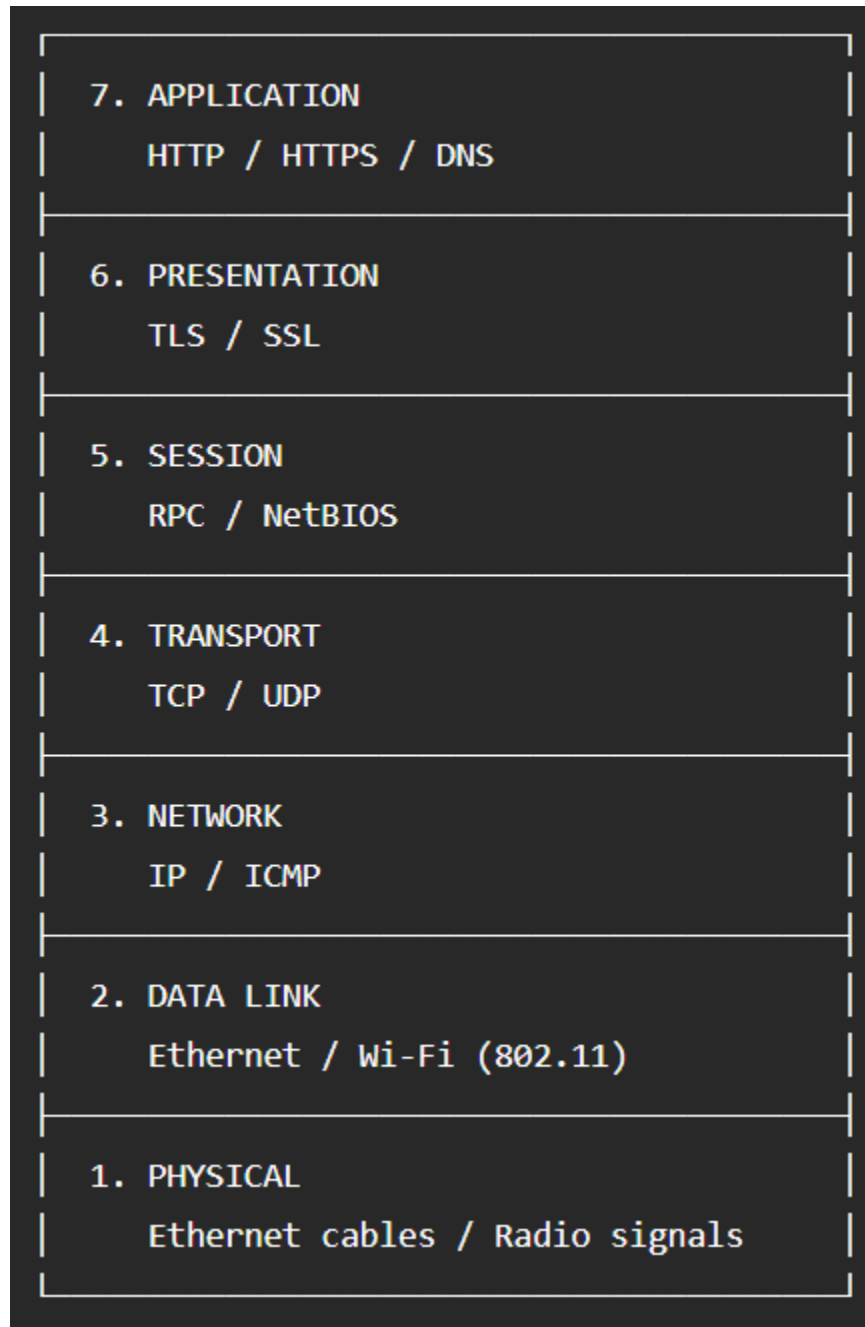


## Task 1: OSI Model – 7 Layers



### Layer Explanation

| Layer | Name | Protocol/Technology | Main Function |
|-------|------|---------------------|---------------|
|-------|------|---------------------|---------------|

|   |              |                       |                                                                                                                 |
|---|--------------|-----------------------|-----------------------------------------------------------------------------------------------------------------|
| 7 | Application  | HTTP/HTTPS, DNS       | Provides network services to applications.                                                                      |
| 6 | Presentation | TLS/SSL               | Handles data representation, encryption, and formatting.                                                        |
| 5 | Session      | RPC, NetBIOS          | Establishes and manages communication sessions.                                                                 |
| 4 | Transport    | TCP, UDP              | Provides end-to-end transport and, with TCP, reliable delivery.                                                 |
| 3 | Network      | IP, ICMP              | Handles logical addressing and routing between networks.                                                        |
| 2 | Data Link    | Ethernet, Wi-Fi       | Handles logical addressing and routing between networks. Handles frames and communication on the local network. |
| 1 | Physical     | Ethernet cable, radio | Transmits raw bits through physical media.                                                                      |

## Task 2: Network Protocols Used by Popular Apps

| App      | Protocol 1 | Protocol 2 | Purpose                                                                                     |
|----------|------------|------------|---------------------------------------------------------------------------------------------|
| WhatsApp | HTTPS      | DNS        | HTTPS is used for secure communication with services, while DNS helps resolve domain names. |

|           |       |     |                                                                                                       |
|-----------|-------|-----|-------------------------------------------------------------------------------------------------------|
| Instagram | HTTPS | DNS | HTTPS protects web/API communication, while DNS helps locate servers.                                 |
| Zomato    | HTTPS | DNS | HTTPS protects requests and responses, while DNS resolves service domains.                            |
| Flipkart  | HTTPS | DNS | HTTPS provides secure application communication, while TCP may provide reliable transport underneath. |
| Paytm     | HTTPS | DNS | HTTPS protects sensitive application communication, while DNS resolves server names.                  |

### Task 3: Ping and Tracert to YouTube

#### Step 1: Ping YouTube

- ping www.youtube.com

#### Step 2: Tracert YouTube

- tracert www.youtube.com

#### OSI Layers Involved

##### Ping

ping primarily uses:

- Layer 3 – Network: ICMP is used to send Echo Request and receive Echo Reply messages.
- Layer 2 – Data Link: The packets must travel across the local network using Ethernet or Wi-Fi frames.
- Layer 1 – Physical: The actual bits travel through cables or wireless radio signals.

##### Tracert

- tracert primarily investigates the Layer 3 routing path.

## Task 4: WhatsApp Message Journey Through the OSI Model

### Scenario

I send the message:

- "Hello, how are you?"
- to a friend using WhatsApp.

The message travels down through the networking layers on my device, across the network, and then back up the layers on my friend's device.

### Sender Side

#### Layer 7 – Application

- I type "Hello, how are you?" in WhatsApp.

The WhatsApp application prepares the message for transmission using its communication protocols and security mechanisms.

#### Layer 6 – Presentation

The data is represented in a suitable format and protected by encryption mechanisms used by the application.

WhatsApp uses end-to-end encryption for personal messages, so the message content is protected from being read by unauthorized parties during transit.

#### Layer 5 – Session

The communication session between the application and its services is maintained so that data can be exchanged.

#### Layer 4 – Transport

Transport protocols handle communication between endpoints.

Depending on the current implementation and network path, modern applications may use TCP-based communication or UDP-based protocols such as QUIC.

### Layer 3 – Network

IP addressing is used to move packets between networks.

The packet travels approximately:

Phone



Wi-Fi Router / Mobile Network



ISP



Internet



WhatsApp Infrastructure



Friend's Network



Friend's Phone

### Layer 2 – Data Link

The data is placed into frames for transmission across the local network, such as Wi-Fi.

The device's network interface uses local network addressing to communicate with the next network device.

### Layer 1 – Physical

The data is transmitted as physical signals, such as radio signals over Wi-Fi/mobile networks or electrical/optical signals elsewhere in the network.